

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to...

- solve multistep linear equations and inequalities in one variable, *naming the property* behind every step;
- decide whether a linear equation has *one*, *no*, or *infinitely many* solutions;
- graph an inequality's solution set on a number line and rearrange a *literal* equation for a specified variable;
- write an equation or inequality from a situation and interpret the answer back in that situation;
- solve a quadratic by square roots, by factoring with the *zero product property*, and by the *quadratic formula*;
- use the *discriminant* to say how many real solutions exist before solving — and match it to the graph.

Vocabulary & Key Concepts

Fill in each term as we build it together.

Equation (vs. expression) / solution set:

Properties of equality:

Identity vs. contradiction:

Literal equation:

Properties of inequality:

Zero product property:

Discriminant:

Hook

Solve all three. Nothing else — just solve them.

(a) $3(x - 2) = x - 6$

(b) $3(x - 2) = 3x - 6$

(c) $3(x - 2) = 3x - 5$

Two of those did something you were not expecting: the variable _____. That is not a mistake — it is the answer arriving in a different costume.

- In (b) you were left with a statement that is _____, so how many values of x work?

- In (c) you were left with a statement that is _____, so how many values of x work?

Now the same question in another family. How many real solutions does each have? $x^2 = 9$: _____
 $x^2 = 0$: _____ $x^2 = -9$: _____

The idea the whole lesson hangs on: an expression was a *recipe*; an equation is a **question**. The answer is a *solution set*, and it is allowed to hold one number, no numbers, or every number. By the end of today you will know the count **before** you solve.

1. Linear Equations — and How Many Answers They Have

An **equation** joins two expressions with $=$. Unlike an expression, it asks a _____, and a **solution** is a value that makes it _____.

Solve, and name every move. The *properties of equality* are the only moves that keep the solution set the same — which is exactly why they are allowed and nothing else is.

Line	Property that makes it legal
$5x + 3(x - 2) = 2(x + 9)$	<i>given</i>
_____	distributive property
$8x - 6 = 2x + 18$	<i>combine like terms</i>
_____	_____
_____	_____
$x =$ _____	_____

Verify it. Substitute your answer into the *original* equation — not into a line partway down, where a mistake would already be baked in.

Left side: _____ Right side: _____ Equal? _____

Checking is not politeness. It is the _____ of the word “solution.”

Back to the Hook: three possible answers. Fill in what the *last line* tells you.

Equation	Last line	Solution set	Name
$3(x - 2) = x - 6$	_____	_____	—
$3(x - 2) = 3x - 6$	_____	_____	_____
$3(x - 2) = 3x - 5$	_____	_____	_____

If the variable survives $\rightarrow$ _____ solution. If it vanishes leaving a **true** statement $\rightarrow$ _____.

If it vanishes leaving a **false** statement $\rightarrow$ _____.

Write one from a situation. A repair shop charges a \$40 flat fee plus \$15 per hour of labor. A bill came to \$130.

Equation: _____ Solve: _____

Now answer the *question that was asked*, in a sentence with units: _____

2. Literal Equations — Same Moves, Scarier Alphabet

A **literal equation** has several letters. “Solve for h ” means: get h _____, using the same properties of equality as always.

Formula	Solve for	Result
$A = \frac{1}{2}bh$ (area of a triangle)	h	_____
$P = 2\ell + 2w$ (perimeter)	w	_____
$y = mx + b$	x	_____
$C = \frac{5}{9}(F - 32)$	F	_____

The one real trap. In $P = 2\ell + 2w$, are 2ℓ and $2w$ like terms? _____ So you may *not* write $P = 4\ell w$. And when you divide, you must divide _____ term, not just the one you were looking at.

Coming attraction: in Unit 2 you will meet $y = mx + b$ rearranged three different ways and hear them called “forms.” They are not three facts — they are one equation, solved for different things.

3. Linear Inequalities — and Why the Symbol Flips

Everything about solving is the same ... except for one move. Let’s find it rather than announce it.

Start with a statement that is definitely true: $2 < 6$.

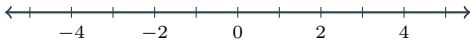
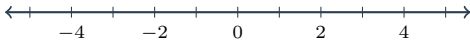
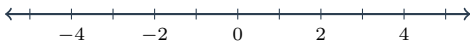
Multiply *both sides* by -1 : -2 _____ -6 (write in the symbol that is actually true)

What happened to the order? _____

On the number line, multiplying by a negative _____ everything across zero — and a reflection _____ the order. So the rule is not a convention somebody chose:

Multiplying or dividing both sides by a negative number _____ **the inequality symbol.** (Adding or subtracting a negative does *not* — only multiplying and dividing.)

Solve and graph. Shade the solution set on the number line. Endpoint circles: *open* $\circ$ for $<$ and $>$, *closed* $\bullet$ for $\leq$ and $\geq$.

Solve	Graph the solution set
(a) $4x - 9 \leq 11$ x _____	
(b) $-3x + 7 > 19$ x _____	
(c) $2(x + 4) \geq 6x - 4$ x _____	

Which one made you flip the symbol, and at exactly which step? — In (c) you may end with $3 \geq x$. Rewrite it as x _____ 3 before you graph — reading the shading off a backwards inequality is how the picture ends up on the wrong side.

Write one from a situation. Concert tickets cost \$12 each and you can spend **at most** \$75.

“At most” translates to the symbol _____. Inequality: _____ Solve: t _____

The algebra says $t \leq 6.25$. How many tickets can you actually buy, and why is the answer not 6.25? _____

4. Quadratic Equations — Three Methods and a Head Count

A **quadratic equation** can be written $ax^2 + bx + c = 0$ with $a \neq$ _____. Which method you use depends on what the equation *looks like*.

Method 1 — Square roots (use when there is no x term). Isolate the square, then take the root of both sides. **Both** roots: the $\pm$ is not optional.

(a) $x^2 = 49$, $x =$ _____ (b) $2x^2 - 50 = 0$, $x =$ _____ (c) $(x - 1)^2 = 16$, $x =$ _____

Writing only $x = 7$ in (a) does not lose a step — it loses _____.

Method 2 — Factor, then the zero product property (use when it factors).

Zero product property: if $mn = 0$ then _____ or _____.
Step 0, every single time: make one side equal _____ first.

Equation	Set to zero, then factor completely	Solutions
$x^2 + 2x - 15 = 0$	_____	_____
$x^2 - 3x = 10$	_____	_____
$2x^2 - 18 = 0$	_____	_____

Why “completely” matters more now. In Lesson 1.1, stopping at $2(x^2 - 9)$ cost you a step. Here it costs you _____.

And why the zero is not negotiable: from $(x - 2)(x + 5) = 8$, may you conclude $x - 2 = 8$? _____ Why not?

Method 3 — The quadratic formula (use when nothing else works — it always works).

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solve $x^2 - 4x - 7 = 0$. $a =$ _____ $b =$ _____ $c =$ _____

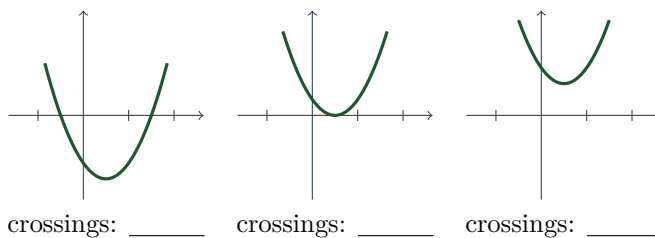
Substitute and simplify:

— these are **irrational** solutions, and simplifying that radical is Lesson 1.1 doing its job.

The head count: the discriminant. The piece under the radical, $b^2 - 4ac$, decides everything *before* you solve. Same a and b below — only c changes.

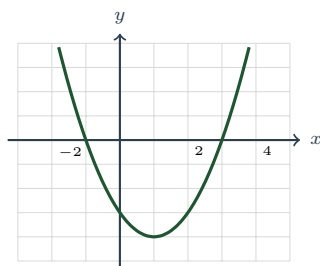
Equation	$b^2 - 4ac$	Positive / zero / negative?	# real solutions
$x^2 - 2x - 3 = 0$	_____	_____	_____
$x^2 - 2x + 1 = 0$	_____	_____	_____
$x^2 - 2x + 3 = 0$	_____	_____	_____

The same three facts, drawn. Below are the graphs of those three, in order. Label each one with the number of times it crosses the x -axis.



The discriminant and the picture are the _____ fact told twice — a real solution *is* an x -intercept.

Read the graph, then confirm with algebra. This is $y = x^2 - 2x - 3$.



Read the x -intercepts off the graph: $x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$

Now confirm by factoring $x^2 - 2x - 3 = 0$:

Do they agree? _____ Finish the sentence: solving $x^2 - 2x - 3 = 0$ is the same task as finding the _____ of the graph.

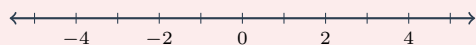
Guided Practice

1. Solve and check: $7(x - 2) = 3x + 6$.

Check: _____

2. How many solutions does $2(3x + 4) = 6x + 8$ have? _____ How do you know?

3. Solve $-5x + 3 \geq 23$ and graph the solution set.



4. Solve for r : $C = 2\pi r$. $r = \underline{\hspace{2cm}}$

5. Without solving, how many real solutions does $x^2 + 4x + 9 = 0$ have? _____ Justify with a number:

6. Solve by factoring: $x^2 + x = 12$.

Verify one solution: _____